

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Pulley alignment fixture, Part Number 93874 (Dayco)

Additional Service Items

- No additional service items required.

Preparatory Steps

⚠ WARNING ⚠

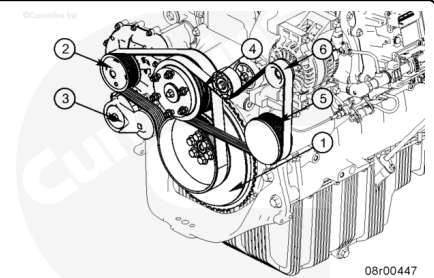
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries. See equipment manufacturer service information.

To be sure the cooling fan drive belt is routed correctly upon installation, make a diagram of the cooling fan belt routing, as shown in the illustration, prior to removing the belt.

The cooling fan belt routing consists of the following components.

1. Crankshaft pulley/vibration damper
2. Water pump pulley
3. Tensioner(s)
4. Fan pulley
5. Refrigerant compressor pulley
6. Alternator pulley.

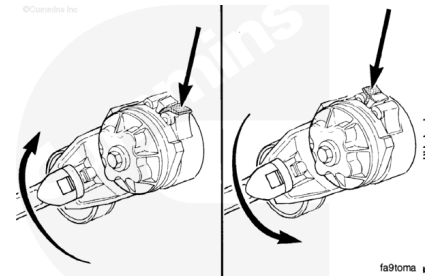


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Remove

⚠ CAUTION ⚠

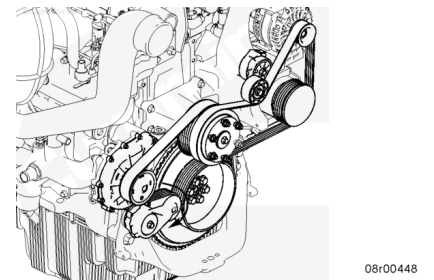
The belt tensioner is spring loaded and must be pivoted away from the drive belt. Pivoting in the wrong direction can result in damage to the belt tensioner or cause personal injury.



The belt tensioner winds in the direction that the spring tang is bent over the tensioner body. To loosen the tension on the belt, rotate the tensioner to wind the spring tighter.

⚠ CAUTION ⚠

Applying excessive force in the opposite direction of windup or after the tensioner has been wound up to the positive stop can cause the tensioner arm to crack or break.



Make a diagram of the belt arrangement prior to removing the drive belt to aid in installation and provide proper routing of the cooling fan drive belt.

Pivot the tensioner in the direction of the spring tang to remove the drive belt.

Remove the drive belt.

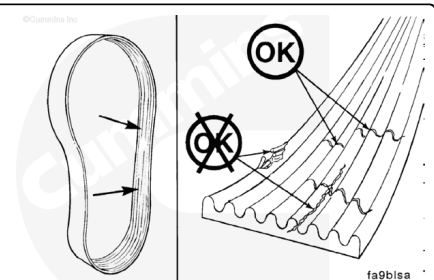
Clean and Inspect for Reuse

Check the belt for damage. Transverse (across the belt width) cracks are acceptable. Longitudinal (direction of the belt length) cracks that intersect with transverse cracks are **not** acceptable. If the belt is frayed or has any piece of material missing, the belt is unacceptable and needs to be replaced.

Inspect the belt for:

- Cracks
- Glazing
- Tears or cuts
- Excessive wear.

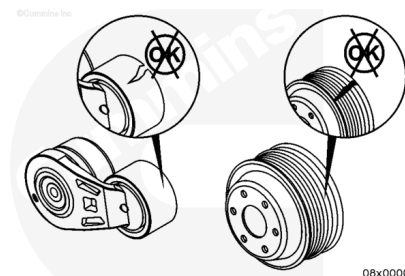
Replace the belt if any damage is found.



Note : If a damaged belt has spun on a pulley, inspect the pulley and assembly for damage. Refer to Procedure 008-089 in Section 8. (</qs3/pubsys2/xml/en/procedures/1016/1016-008-089-tr.html>)

Inspect the pulleys for wear or cracks.

Inspect the belt tensioner. Refer to Procedure 008-087 in Section 8. (</qs3/pubsys2/xml/en/procedures/1016/1016-008-087-tr.html>)



⚠ CAUTION ⚠

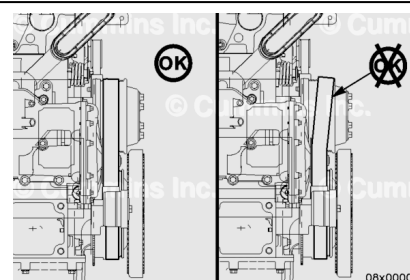
A worn belt tensioner or mis-aligned pulley can cause damage to the belt.

Identify misalignment of pulleys down to $\frac{1}{2}$ of a belt rib. The capability of the alignment fixture is contingent upon proper calibration. Verify all set-up and calibration procedures are followed before verifying pulley alignment.

Verify all pulleys are aligned correctly using the pulley alignment fixture tool or equivalent:

- Part Number 93874 (Dayco).

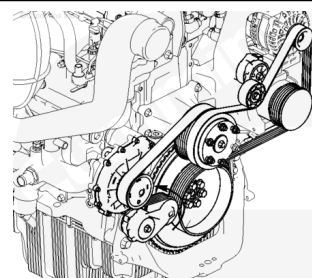
Pulley alignment should **always** be checked from the fan pulley, and crankshaft pulley to other front end mounted grooved pulleys. Contact a Cummins® authorized repair location for drive belt alignment.



Install

⚠ CAUTION ⚠

The belt tensioner is spring loaded and must be pivoted away from the drive belt. Pivoting in the wrong direction can result in damage to the belt tensioner or cause personal injury.



⚠ CAUTION ⚠

To reduce the possibility of damage to the pulley and new belt, do not roll the belt over the pulley or pry on the belt with a tool.

Use the belt diagram created in the Remove section to route the drive belt on the engine.

Install a new fan belt on the pulleys.

Pivot the tensioner in the direction of the spring tang and install the drive belt, slipping the belt over the water pump pulley last.

Release the tensioner to apply tension to the drive belt.

Check the alignment of the belt with the tensioner and the rest of the front-end auxiliary drive.

Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Connect the batteries. See equipment manufacturer service information.
- Operate the engine and check for belt squeal. Excessive belt squeal indicates belt slippage.
- If belt squeal is present, check the routing of the belt and verify that the belt is installed correctly on each pulley.